

SWGITM — Demo to Production Readiness Directive

Objective

Transition SWGITM from Google sandbox demo into a production-grade, OpenShift-deployable container stack suitable for Red Hat marketplace evaluation and enterprise deployment.

This directive defines required architecture hardening, packaging standards, lifecycle compliance, and deployment expectations.

Phase 1 — Lock and Tag Google Sandbox

- 1 . Finalize Google demo functionality.
- 2 . Ensure deterministic receipt generation is validated.
- 3 . Confirm server-side receipt persistence.
- 4 . Confirm receipt retrieval by `receipt_id` via API.
- 5 . Tag release: `google-sandbox-v1.0.0`
- 6 . Freeze sandbox repo except for bug fixes.

Sandbox remains demo environment. Production work proceeds separately.

Phase 2 — Extract SWGI Core

Create clean boundary between demo layer and core engine.

SWGI Core Must Contain:

- Policy evaluation engine
- Authorization gate logic
- Deterministic decision engine
- Trust Receipt generation + hashing
- Receipt schema validation
- Receipt store interface abstraction

Core must NOT include:

- UI code
- Demo workflows

- GCP-specific logic
- Hardcoded configuration

Persistence Abstraction

Core must call a storage interface such as:

```
ReceiptStore.write(receipt)
ReceiptStore.get(receipt_id)
ReceiptStore.query(filters)
```

Implementations:

- Sandbox → SQLite adapter
- Production → Postgres adapter

Core must remain storage-agnostic.

Phase 3 — Production Repository Creation

Create new repository: `swgi-openshift`

This repository represents the OpenShift production container stack.

Import SWGI Core into this repo.

This repo must contain:

- Production API service
 - Postgres backend integration
 - Container definitions
 - Helm chart or Operator
 - RBAC definitions
 - Documentation
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Phase 4 — Production Backend Requirements

Database

- Postgres required (containerized or managed)
- Restart-safe persistence

- No SQLite in production
- Migration strategy defined

API Standards

- Versioned API (/v1/ . . .)
- OpenAPI/Swagger specification
- Auth required (JWT or service token)
- Role-based access (admin/viewer)

Minimum endpoints:

- POST /authorize
 - GET /receipts/{receipt_id}
 - GET /receipts
 - GET /health
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Phase 5 — Container Hardening

Each container must:

- Use minimal base image (UBI or distroless)
- Run as non-root user
- Expose explicit ports only
- Define HEALTHCHECK
- Use semantic version tags
- Pass vulnerability scan (no critical CVEs)

Push images to registry compatible with Red Hat review.

Phase 6 — OpenShift Packaging

Required:

- Helm chart OR OpenShift Operator
- Deployment
- Service
- ConfigMap
- Secret
- ServiceAccount
- Role

- RoleBinding
- Resource requests & limits defined

RBAC

- Least privilege only
 - No cluster-admin usage unless strictly required
 - Dedicated namespace deployment
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Phase 7 — Lifecycle Compliance

SWGf must support:

- Clean install
- Clean upgrade
- Clean uninstall
- No orphaned resources
- Versioned releases
- Changelog

Helm install, upgrade, and uninstall must execute without error.

Phase 8 — Observability & Security

Required:

- Structured logging (JSON preferred)
- Log levels configurable
- Liveness probe
- Readiness probe
- TLS support
- Secrets via Kubernetes Secret
- No hardcoded credentials

Optional but recommended:

- Prometheus metrics endpoint
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Phase 9 — Production Mode Controls

Define two modes:

Sandbox Mode

- SQLite allowed
- Minimal security
- Demo-focused

Production Mode

- Postgres required
- TLS enforced
- Auth required
- RBAC active

Mode must be environment-driven.

Phase 10 — Documentation Package

Production repo must include:

- Architecture diagram
 - Install guide
 - Upgrade guide
 - RBAC explanation
 - Support contact
 - Known limitations
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Final Outcome

SWGI production release must be:

- Containerized
- OpenShift-deployable
- Cloud-agnostic
- GCP-built
- Postgres-backed
- Deterministic

- Receipt-verifiable
- Enterprise lifecycle compliant

This release becomes the official OpenShift product channel while the Google sandbox remains demonstration vehicle.

End of directive.